

MINGRUI ZHANG

Research Engineer
Exa, Inc
San Francisco, CA, USA 94110

Email: z1m6r3@gmail.com
<https://www.drustz.com>

EDUCATION

University of Washington, Seattle

Sep. 2017 - June 2022

Ph.D. in Information Science, The Information School

Focus: Human-Computer Interaction

Advisor: Jacob O. Wobbrock

Tsinghua University, Beijing

Aug. 2013 - Jul. 2017

B.Eng. with Honors, Department of Computer Science and Technology

Minor in journalism

The Chinese University of Hong Kong, Hong Kong

Sep. 2015 - Dec. 2015

Exchange student, Department of Computer Science and Engineering

RESEARCH INTEREST

Large Language Models, Agentic AI and Human-Computer Interaction

My research interests center on Agentic AI and Large Language Models, focusing on 1) building LLM-based agents capable of reasoning, planning, and interacting with complex environments — search, tools, terminals — much like humans do; 2) advancing AI-mediated interaction paradigms that make intelligent systems more intuitive, communicative, and human-like companions.

PROFESSIONAL EXPERIENCE

Reflection.ai. San Francisco, CA

Aug. 2026 - Present

Research Engineer

Building Superknowledge.

Reflection.ai. New York, NY

Nov. 2025 - Jul. 2026

Member of Technical Staff

- LLM-inference data pipeline* Built and owned the mid-training data curation pipeline with GPU clusters for asynchronous inference, transforming in-house and open-source raw data into 2M+ high-quality data pairs via automated rewriting and synthetic data generation.

- Math/STEM data curation* Work with vendors to co-design the quality graders, answer verification and task difficulty identification frameworks. Training model to debug the eval trace quality and iterative on the curation-training loop.

- Internal tooling* Founded 3 internal ML tools (trace debugging, eval comparison, data inspection / visualization) adopted as core training infrastructure across the 80+ person modeling org; built agentic workflows for ML observability and data onboarding monitoring.

Meta Platforms Inc. New York, NY

Oct. 2024 - Nov. 2025

Modern Recommendation Systems of Meta AI

Research Scientist

- User understanding, cohort modeling, and interest embedding for Instagram and Facebook.

- Model architecture design for LLM based contextual embedding for user biography, integrated with Meta's HSTU generative recommendation model.

- Post-training (SFT + RL) of ranking LLM agents, including agent orchestration and evaluation data curation for internal ranking tasks. Achieved 1st place on TableBench benchmark for table understanding with our RankAgent model.

Meta Reality Labs. New York, NY

Oct. 2022 - Oct. 2024

Input Exploration Team

Research Scientist

- Research lead for text input interaction on AR glasses (Meta Rayban Display) with gestural interface based on EMG signals, including keyboard, conversational dictation, and EMG handwriting.
- Input interaction prototyping with hand + gaze for next generation mix reality headset (Quest).

Apple Inc. Seattle, WA

Jun. 2021 - Sep. 2021

Advised by Huy Viet Le and Tim Paek

Research Intern

- Gaze-based dictation interaction look to dictate/edit for Apple Vision Pro, including interface prototype and user intent classification modeling.

Facebook Reality Labs. New York, NY

Jun. 2020 - Oct. 2020

Advised by Adam Berenzweig

Research Intern

- EMG-based QWERTY-style text entry interactions, enabling ten-finger typing on any surface.
- Enhanced neural network structure and data collection pipeline with CTC loss for gesture classification.

Google Inc. Mountain View, CA

Oct. 2019 - Dec. 2019

Advised by Shumin Zhai

Research Intern

- Sentence level Gboard auto-correction features, see publication ??.

Momenta.ai Inc. Beijing

Dec. 2016 - Jun. 2017

Advised by Ji Liang

Research Intern

- Enhanced road segmentation models, increasing recall and accuracy by 5%.
- Optimized framework speed from 8fps to 100fps.

MailTime Inc. Beijing

Feb. 2016 - Jun. 2016

iOS Intern

- Designed and implemented the iOS interaction logic and UI, improving user retention by 25%.

Chestnut Tech Inc. Beijing

Oct. 2014 - Jan. 2016

Co-founder, iOS Developer

- Developed “Parocam” application on iOS, focusing on face transform algorithms and UI design.

SELECTED PEER-REVIEWED CONFERENCE PAPERS

For the full publication list, go to my google scholar page

C.5 Yuhang Zhou, **Mingrui Zhang**, Ke Li, Mingyi Wang, Qiao Liu, Qifei Wang, Jiayi Liu, Fei Liu, Serena Li, Weiwei Li, Mingze Gao, Abhishek Kumar, Xiangjun Fan, Zhuokai Zhao, Lizhu Zhang. (2026) Mixture-of-Minds: Multi-Agent Reinforcement Learning for Table Understanding. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL '26)*.

C.4 Patrick Kaifosh, Thomas R. Reardon & CTRL-labs at Reality Labs (including **Mingrui Zhang**) (2025) A generic non-invasive neuromotor interface for human-computer interaction. ***Nature***.

C.3 **Mingrui “Ray” Zhang**, Shumin Zhai, Jacob O. Wobbrock. (2022) TypeAnywhere: A QWERTY-Based Text Entry Solution for Ubiquitous Computing. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '22)*. New Orleans, USA (April 30 - May 6, 2022). New York: ACM Press. [25%]

C.2 **Mingrui “Ray” Zhang**, He Wen, Jacob O. Wobbrock. (2019). Type, Then Correct: Intelligent Text Correction Techniques for Mobile Text Entry Using Neural Networks. *Proceedings of the*

32nd Annual ACM Symposium on User Interface Software & Technology (UIST '19). New York: ACM Press. ^[24%]

C.1 Xin Yi, Chun Yu, **Mingrui “Ray” Zhang**, Sida Gao, Ke Sun, Yuanchun Shi. (2015). ATK: Enabling Ten-Finger Freehand Typing in Air Based on 3D Hand Tracking Data. *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (UIST '15)*. New York: ACM Press. ^[24%]

HONORS & AWARDS

Ford Fellowship, University of Washington	<i>2022</i>
Best Paper Award, ACM CHI	<i>2019</i>
Excellent Graduate of the CST Department, Tsinghua	<i>2017</i>
The National Scholarship of China (Top 1%)	<i>2016</i>
1st Winner of the National Database Conference Cup	<i>2016</i>